

Chapter 48: Pharmacogenomics and Personalized Medicine: The Future of Precision Therapeutics

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Abstract

Pharmacogenomics has emerged as a cornerstone of personalized medicine by elucidating how genetic variations influence drug efficacy, safety, metabolism, and dosage requirements. Conventional therapeutic approaches based on population averages often fail to account for individual differences in treatment response, leading to suboptimal outcomes and adverse drug reactions. Pharmacogenomics addresses this challenge by identifying genetic determinants involved in pharmacokinetics, pharmacodynamics, drug transport, immune-mediated toxicity, and disease-specific therapeutic response. This chapter provides a comprehensive overview of the scientific principles, historical evolution, and clinical applications of pharmacogenomics in precision therapeutics. It highlights the role of clinically significant biomarkers and genotype-guided therapy in oncology, cardiovascular medicine, psychiatry, pain management, infectious diseases, and immunology. Emerging advances in next-generation sequencing, artificial intelligence, and multi-omics integration are expected to expand the role of pharmacogenomics, making precision therapeutics an essential component of future healthcare and improving patient outcomes worldwide.

Keywords: *pharmacogenomics, personalized medicine, precision therapeutics, drug response, genetic polymorphism, biomarkers, companion diagnostics, precision oncology*

1. Introduction

The history of therapeutics is marked by a persistent tension between standardization and individualization. Classical pharmacology has generated essential principles governing the actions, disposition, and safety of drugs, yet therapeutic decisions in routine practice have long been guided by average responses derived from population-based evidence. Although this approach has transformed clinical care, it does not adequately explain why one patient benefits remarkably from a drug, another derives little effect, and a third develops severe toxicity despite receiving the same agent at the same dose. Such variability reflects the complex interplay of age, sex, organ function, diet, comorbid disease, concomitant medication use, environmental exposures, epigenetic regulation, and genomic background. Among these determinants, genetic variation has become especially important because it offers a mechanistically grounded and increasingly measurable basis for predicting drug response.

Pharmacogenomics, broadly defined, is the study of the influence of the genome on drug efficacy, toxicity, metabolism, and dosing requirements. While the older term *pharmacogenetics* often referred to single-gene effects on drug response, pharmacogenomics encompasses genome-wide influences, multigene interactions, transcriptomic signatures, somatic alterations, and systems-level interpretation. In practice, the distinction between the two terms is often blurred, but pharmacogenomics better reflects the scale and scope of contemporary precision therapeutics. Personalized medicine, sometimes used interchangeably with precision medicine, refers to tailoring prevention, diagnosis, and treatment according to biologic and contextual features of the individual patient. Within this broader framework, pharmacogenomics serves as one of the most clinically actionable tools for treatment optimization.

The importance of pharmacogenomics lies not merely in the possibility of refining therapy, but in its potential to alter the logic of prescribing itself. Rather than reacting to treatment failure or adverse effects after exposure has occurred, genomic-informed medicine seeks to anticipate risk and match the right drug to the right patient at the right dose and at the right time. In certain areas, especially oncology, this paradigm has already transformed clinical outcomes through the use of targeted agents selected according to molecular characteristics of tumors. In other settings, such as cardiovascular medicine, psychiatry, pain management, and infectious diseases, implementation remains heterogeneous but is steadily expanding as evidence accumulates and testing becomes more accessible. The future of precision therapeutics will likely depend on integrating pharmacogenomic insights with other layers of biological and clinical data. A patient's genotype alone rarely provides a complete answer. Drug response is shaped by dynamic physiology, disease heterogeneity, microbiome composition, inflammatory state, concurrent therapy, and behavioral factors such as adherence. Thus, the most effective model of personalized medicine is not genetic determinism but informed integration. Pharmacogenomics should therefore be understood as a high-value component of individualized therapy rather than an isolated predictive instrument.[1]

This chapter presents a comprehensive overview of pharmacogenomics and its role in personalized medicine. It explores the scientific foundations of genomic variation in drug response, major clinically relevant biomarkers, therapeutic applications across specialties, implementation challenges, and future directions that may define the next generation of precision therapeutics.

2. Historical Evolution of Pharmacogenomics and Personalized Medicine

The intellectual roots of pharmacogenomics can be traced to observations made long before the era of molecular genetics. Clinicians had long recognized unusual patterns of drug sensitivity in certain individuals and families. Early examples included prolonged apnea after succinylcholine administration, hemolysis in glucose-6-phosphate dehydrogenase deficient individuals exposed to oxidant drugs, and peripheral neuropathy associated with isoniazid in slow acetylators. These phenomena established that inherited traits could profoundly influence drug disposition and toxicity. However, the field remained limited by the lack of tools to identify specific molecular mechanisms.

The emergence of molecular biology, human genetics, and recombinant DNA technology in the latter half of the twentieth century transformed this landscape. Variants in genes encoding drug-metabolizing enzymes, particularly those in the cytochrome P450 superfamily, were progressively characterized and linked to clinically meaningful differences in metabolism. The discovery of polymorphisms in CYP2D6, CYP2C19, and thiopurine S-methyltransferase provided some of the earliest compelling evidence that genotype-informed prescribing could improve safety and efficacy. [2]

The completion of the Human Genome Project marked a decisive turning point. It accelerated the cataloging of single nucleotide polymorphisms, copy number variants, insertions, deletions, and structural genomic alterations, enabling broader association studies and more systematic biomarker discovery. Parallel advances in high-throughput genotyping, next-generation sequencing, transcriptomics, and bioinformatics expanded pharmacogenomics from candidate-gene analysis to genome-wide approaches. The field no longer focused solely on inherited differences in metabolism, but also on somatic mutations, gene expression profiles, pathway activation states, and immune-related susceptibility to adverse drug reactions.

At the same time, medicine itself was undergoing a conceptual shift. The rise of evidence-based medicine had emphasized standardized protocols based on clinical trials, but the growing recognition of

disease heterogeneity exposed the limitations of one-size-fits-all strategies. Cancers once classified only by tissue of origin were subdivided by molecular drivers. Psychiatric conditions and inflammatory disorders showed variable response patterns that could not be fully predicted by clinical phenotype alone. Personalized medicine thus emerged as a response to both therapeutic uncertainty and biological complexity.

In recent years, the phrase *precision medicine* has gained prominence because it implies stratification based on measurable characteristics rather than the unrealistic notion that every treatment can be custom designed for each individual. Nevertheless, in practical discourse, personalized medicine remains widely used and continues to convey the patient-centered goal of tailoring care. Pharmacogenomics occupies a central place in this evolution because it supplies one of the most mature and actionable forms of molecular stratification in therapeutics.

3. Conceptual Basis of Pharmacogenomics

Drug response can be understood through two broad dimensions: pharmacokinetics and pharmacodynamics. Pharmacokinetics describes what the body does to a drug, including absorption, distribution, metabolism, and excretion. Pharmacodynamics describes what the drug does to the body, including receptor binding, signal transduction, downstream biological effects, therapeutic outcomes, and adverse effects. Genetic variation can influence both domains, thereby altering systemic exposure, target engagement, and clinical response.

Pharmacokinetic variability is often driven by polymorphisms in genes encoding phase I enzymes, phase II conjugating enzymes, transporters, and proteins involved in drug uptake and elimination. For example, reduced-function alleles in CYP2C19 can impair the activation of clopidogrel, leading to diminished antiplatelet effect, while variants in CYP2D6 can classify patients as poor, intermediate, normal, or ultrarapid metabolizers for numerous antidepressants, antipsychotics, opioids, and antiarrhythmics. Similarly, reduced activity of thiopurine S-methyltransferase or nudix hydrolase 15 can increase exposure to active thiopurine metabolites, predisposing patients to life-threatening myelosuppression.

Pharmacodynamic variability may arise from differences in receptors, enzymes, signaling mediators, ion channels, immune recognition molecules, or disease-related molecular targets. Variants in VKORC1 influence sensitivity to warfarin, whereas certain HLA alleles are strongly associated with severe hypersensitivity reactions to specific drugs. In oncology, somatic alterations such as EGFR mutations, ALK rearrangements, BCR-ABL fusion, HER2 amplification, or BRAF mutations determine responsiveness to targeted agents. These examples illustrate that pharmacogenomics is relevant not only to inherited host genetics but also to the molecular characteristics of diseased tissue.

A further dimension is the genomic basis of idiosyncratic and immune-mediated adverse drug reactions. Such reactions often occur unpredictably and may be severe despite standard dosing. Associations between HLA genotypes and adverse reactions, such as HLA-B57:01 with *abacavir hypersensitivity* or HLA-B15:02 with carbamazepine-induced Stevens-Johnson syndrome in certain populations, demonstrate that genomic screening can sometimes prevent catastrophic toxicity. [3]

In contemporary therapeutics, pharmacogenomics is best conceptualized as a framework for predictive enrichment. It identifies those most likely to respond, those who require dose adjustment, and those at high risk for harm. The value of this framework increases when genomic information is interpreted alongside clinical variables. For instance, warfarin dose requirements depend on CYP2C9 and VKORC1 genotype, but also on age, diet, liver function, interacting drugs, and body size. Similarly,

somatic mutation status in cancer must be interpreted within the broader context of tumor burden, resistance pathways, prior therapy, and performance status.

4. Genetic Determinants of Drug Response

Variants in Drug-Metabolizing Enzymes: The cytochrome P450 system remains one of the most intensively studied domains in pharmacogenomics. Enzymes such as CYP2D6, CYP2C19, CYP2C9, CYP3A5, and CYP2B6 exhibit substantial interindividual genetic variability. These polymorphisms can lead to absent, reduced, normal, or increased metabolic activity, which in turn affects plasma drug concentration and therapeutic outcome. CYP2D6 is particularly important because it metabolizes a large number of centrally acting drugs, beta-blockers, antiarrhythmics, and opioid prodrugs. Individuals with nonfunctional alleles may experience drug accumulation and toxicity when treated with substrates requiring inactivation, whereas ultrarapid metabolizers may have subtherapeutic exposure. For prodrugs such as codeine, the opposite pattern may occur: poor metabolizers fail to generate sufficient active morphine, while ultrarapid metabolizers may produce excessive amounts, increasing toxicity risk. CYP2C19 affects proton pump inhibitors, certain antidepressants, antifungal agents, and clopidogrel. Loss-of-function alleles reduce clopidogrel bioactivation and have been associated with poorer platelet inhibition. CYP2C9 plays a major role in metabolizing warfarin and several nonsteroidal anti-inflammatory drugs. Reduced-function alleles are clinically significant because they can prolong drug exposure and amplify bleeding risk with warfarin or toxicity with other narrow therapeutic index drugs. Among phase II enzymes, thiopurine S-methyltransferase and UDP-glucuronosyltransferases deserve particular attention. Thiopurine S-methyltransferase deficiency can cause severe myelotoxicity when standard doses of azathioprine, mercaptopurine, or thioguanine are used. UGT1A1 polymorphisms influence glucuronidation capacity and are relevant to irinotecan toxicity as well as bilirubin metabolism.

Drug Transporters: Drug transporters regulate cellular uptake, efflux, tissue distribution, and elimination. Genes such as SLCO1B1, ABCB1, ABCG2, and SLC22A1 contribute to clinically relevant variability. SLCO1B1 encodes the hepatic uptake transporter OATP1B1, and reduced-function variants can increase circulating concentrations of statins, particularly simvastatin, thereby raising the risk of myopathy. Transporter polymorphisms may also influence penetration into sanctuary sites such as the blood-brain barrier and thus modify efficacy in central nervous system disorders. [4]

Drug Targets and Signaling Pathways: Pharmacodynamic determinants include polymorphisms in receptors, enzymes, ion channels, and downstream signaling molecules. VKORC1 variants affect warfarin sensitivity by altering the target enzyme of vitamin K antagonism. Beta-adrenergic receptor polymorphisms have been investigated in relation to beta-blocker response, although clinical implementation remains limited due to inconsistent evidence. In oncology, the relevance of target variation is more direct and often profound. Tumor cells bearing activating EGFR mutations, for instance, may be exquisitely sensitive to certain tyrosine kinase inhibitors, whereas KRAS mutations predict lack of benefit from specific anti-EGFR monoclonal antibodies in colorectal cancer.

Human Leukocyte Antigen and Immune-Mediated Toxicity: HLA genotypes are among the strongest pharmacogenomic predictors of severe adverse drug reactions. The immune system can recognize drug-derived antigens or altered self-peptides in an HLA-restricted manner, leading to hypersensitivity syndromes that range from cutaneous reactions to systemic organ injury. Screening for HLA-B*57:01 before abacavir therapy is a landmark example of successful clinical implementation, as it markedly reduces hypersensitivity risk. Other HLA-drug associations have informed safer prescribing in selected populations, especially where allele prevalence and reaction severity justify pre-treatment testing.

5. Technologies Enabling Pharmacogenomics

The clinical growth of pharmacogenomics has been made possible by rapid advances in genomic technologies. Early assays focused on individual variants using polymerase chain reaction-based methods, restriction fragment analysis, or allele-specific probes. These approaches remain valuable for targeted testing where a small number of highly actionable variants is sufficient. However, the field has increasingly moved toward panel-based genotyping, microarrays, and next-generation sequencing. [5]

Targeted pharmacogenomic panels offer practical advantages because they can test multiple clinically relevant genes simultaneously at relatively low cost and with manageable interpretive complexity. Such panels are often used preemptively so that genomic information is available at the point of prescribing. Whole-exome and whole-genome sequencing broaden the scope of discovery and can reveal rare variants, copy number variation, and structural alterations, but they also generate significant analytic and ethical challenges, including incidental findings and uncertain functional interpretation.

In oncology, tissue-based molecular profiling has become integral to treatment selection. Techniques include targeted sequencing, fluorescence in situ hybridization, immunohistochemistry, gene expression profiling, and liquid biopsy. Circulating tumor DNA analysis is especially promising because it permits dynamic monitoring of tumor evolution, treatment response, minimal residual disease, and acquired resistance mutations without repeated invasive biopsy.

Beyond genomics, the future of precision therapeutics increasingly depends on multi-omics integration. Transcriptomics, proteomics, metabolomics, epigenomics, and microbiome analysis can reveal functionally relevant layers that genotype alone may miss. Artificial intelligence and advanced bioinformatics are being used to synthesize these complex datasets and generate clinically interpretable predictions. Nonetheless, the quality of prediction depends heavily on data diversity, annotation accuracy, and prospective validation.

6. Clinical Applications of Pharmacogenomics

Pharmacogenomics has significantly transformed clinical practice by enabling personalized therapeutic strategies based on an individual's genetic profile. Its application across diverse medical specialties enhances drug efficacy, minimizes adverse drug reactions, and supports evidence-based precision medicine through genotype-guided treatment decisions. [6]

Oncology: Oncology has become the flagship domain of precision therapeutics because many cancers are driven by definable molecular abnormalities that can be matched to targeted agents. Examples include EGFR-mutated non-small cell lung cancer treated with EGFR inhibitors, HER2-positive breast cancer treated with anti-HER2 therapies, BCR-ABL-positive chronic myeloid leukemia treated with tyrosine kinase inhibitors, and BRAF-mutant melanoma treated with BRAF and MEK inhibitors. These therapeutic successes illustrate a central principle of pharmacogenomics in oncology: treatment efficacy often depends more on molecular architecture than on tissue histology alone. Companion diagnostics are critical in this setting. They identify the presence or absence of actionable biomarkers and guide drug eligibility. In addition to improving response rates, this strategy reduces unnecessary exposure to ineffective therapies and helps rationalize cost. Modern oncology also increasingly recognizes resistance evolution. Tumors may initially respond to targeted treatment but later acquire secondary mutations, bypass pathway activation, phenotypic transformation, or intratumoral clonal diversification. As a result, pharmacogenomics in oncology is dynamic rather than static, requiring serial molecular reassessment. Immuno-oncology adds another layer to personalized medicine. Biomarkers such as microsatellite instability, tumor mutational burden, PD-L1 expression, and mismatch repair deficiency can influence the

probability of response to immune checkpoint inhibitors. Although these markers are imperfect, they reflect the expanding concept of pharmacogenomics beyond traditional inherited polymorphisms to include molecular determinants of therapeutic susceptibility within tumors and the tumor microenvironment.

Cardiovascular Medicine: Cardiovascular pharmacogenomics has focused strongly on antiplatelet therapy, anticoagulation, and lipid-lowering agents. Clopidogrel requires metabolic activation, and reduced-function CYP2C19 alleles can decrease active metabolite formation, with potential implications for antiplatelet efficacy in selected clinical settings. Warfarin dosing is influenced by variants in CYP2C9 and VKORC1, along with clinical factors such as age, body size, diet, and interacting medications. Genotype-guided dosing has shown value in refining initial dose estimation, though implementation varies across health systems and is influenced by the availability of alternative anticoagulants. SLCO1B1 genotyping has relevance in statin therapy, particularly for identifying higher risk of simvastatin-associated myopathy. Although not all patients require routine testing, this is a useful example of how pharmacogenomics can be applied not only to maximize efficacy but also to reduce toxicity and improve treatment adherence.

Psychiatry: Psychiatric pharmacotherapy is characterized by delayed response assessment, frequent trial-and-error prescribing, variable tolerability, and substantial nonadherence. These features make psychiatry an attractive domain for pharmacogenomics. Variants in CYP2D6 and CYP2C19 influence the metabolism of many antidepressants and antipsychotics, and genotype-informed dose adjustment can sometimes reduce adverse effects or improve drug selection. However, the field remains more complex than simple single-gene prediction. Psychiatric outcomes are influenced by diagnosis heterogeneity, placebo effects, psychosocial factors, comorbidity, and polypharmacy. Consequently, pharmacogenomics in psychiatry is best viewed as a tool for therapeutic refinement rather than a stand-alone determinant of treatment choice.

Pain Medicine: Pain management offers several clinically relevant pharmacogenomic examples. CYP2D6 status affects the activation of codeine and tramadol. Poor metabolizers may derive inadequate analgesia, while ultrarapid metabolizers may be at greater risk of opioid toxicity. Genetic variation may also influence non-opioid analgesic response and susceptibility to adverse events, although many proposed associations still require stronger validation.

Infectious Diseases: In infectious diseases, pharmacogenomics has improved both safety and efficacy. HLA-B*57:01 screening before abacavir use has become a standard example of precision prescribing preventing severe hypersensitivity. Variants affecting drug transport, metabolism, and mitochondrial toxicity have also been explored in antiretroviral therapy. Host genetic factors can influence antiviral responses more broadly, though the clinical utility of specific tests varies as treatment regimens evolve.

Immunology and Autoimmune Disorders: Biologics and small-molecule targeted therapies used in rheumatology, gastroenterology, and dermatology have expanded opportunities for personalized treatment. Research is exploring genomic and transcriptomic predictors of response to tumor necrosis factor inhibitors, interleukin antagonists, Janus kinase inhibitors, and other immunomodulatory agents. Although implementation is not yet routine in many of these fields, the trajectory suggests increasing use of molecular stratification to reduce ineffective treatment cycles and optimize sequencing of advanced therapies.

Pharmacogenomics and Adverse Drug Reactions: Adverse drug reactions remain a major cause of morbidity, hospitalization, and healthcare cost worldwide. Many are predictable extensions of

pharmacologic action, but others are disproportionate, severe, or seemingly idiosyncratic. Pharmacogenomics offers one of the most promising strategies for prevention, particularly when strong genotype-phenotype relationships exist. Toxicity may arise from excessive exposure due to impaired metabolism, altered transport, or reduced elimination. Examples include thiopurine-induced myelosuppression in patients with low thiopurine methylation capacity, fluoropyrimidine toxicity in individuals with dihydropyrimidine dehydrogenase deficiency, and irinotecan-associated neutropenia linked to reduced UGT1A1 activity. In such cases, pre-treatment testing can facilitate dose reduction or alternative therapy. Immune-mediated reactions represent another major area of impact. Severe cutaneous adverse reactions, drug-induced liver injury, and systemic hypersensitivity syndromes may have strong HLA associations, though penetrance is often incomplete. The utility of testing depends on several factors: severity of the adverse event, prevalence of the allele, availability of safer alternatives, and predictive performance of the biomarker in the relevant population. A key implication is that pharmacogenomics can shift drug safety from passive surveillance to active prevention. Instead of identifying susceptible individuals only after harm occurs, clinicians can prospectively stratify risk. This has important ethical and economic implications, especially for drugs with serious but preventable toxicities.

7. Personalized Medicine Beyond Pharmacogenomics

Although pharmacogenomics is central to precision therapeutics, personalized medicine extends further. Optimal treatment decisions often require the integration of genomic data with clinical phenotype, imaging, pathology, environmental exposure, microbiome composition, patient preference, and digital health information. A purely gene-centric approach risks oversimplification because the same genotype may produce different outcomes in different physiologic or therapeutic contexts. [7]

For example, a patient's response to a drug may depend on inflammation-induced phenoconversion, in which non-genetic factors suppress enzyme activity and functionally convert a genotypic normal metabolizer into a phenotypic poor metabolizer. Similarly, drug-drug interactions can profoundly alter pharmacokinetics regardless of genotype. Renal and hepatic function, age-related changes in volume of distribution, nutritional status, and adherence all contribute meaningfully to therapeutic variability.

Precision medicine therefore requires layered interpretation. Genomics may identify predisposition, but clinical judgment determines relevance. The future lies not in replacing bedside reasoning with molecular algorithms, but in enriching clinical decision-making with more precise biological information. This integrative model is particularly important in complex chronic disease, where treatment goals, comorbidity burden, and patient values shape the choice of therapy as much as molecular eligibility does.

8. Emerging Frontiers in Precision Therapeutics

The future of pharmacogenomics will likely be shaped by several converging developments. [8]

- One is the movement from single-gene testing toward polygenic and multimodal prediction. Many drug responses are influenced by numerous variants of modest effect rather than a single major determinant. Polygenic models, when combined with clinical covariates, may improve prediction for complex traits such as antidepressant response, analgesic sensitivity, and susceptibility to common adverse effects.

- A second frontier is dynamic molecular monitoring. Liquid biopsy in oncology already demonstrates how therapy can be adapted in near real time according to evolving tumor genomics. Similar principles may eventually extend to transplant medicine, infectious disease, and immune monitoring.
- A third development is the integration of pharmacogenomics with artificial intelligence. Machine learning systems can identify high-order interactions among genes, comedications, laboratory trends, and clinical outcomes that are difficult to capture with traditional models. However, algorithmic transparency, interpretability, and bias control are critical, especially in high-stakes therapeutic decision-making.
- A fourth frontier involves rare variant interpretation. Much of classical pharmacogenomics has focused on common alleles, but rare functional variants may collectively account for substantial unexplained variability. Large-scale sequencing linked to deeply phenotyped clinical cohorts will be important for uncovering these effects.
- A fifth trend is the expansion of precision therapeutics beyond specialist centers into primary care and population health. Preemptive pharmacogenomic profiling may eventually become part of longitudinal health records, much like blood group or allergy status. If implemented responsibly, this could shift pharmacogenomics from a niche intervention to a routine component of prescribing.

9. Conclusions

Pharmacogenomics is transforming modern therapeutics by enabling genotype-guided drug selection and dose optimization, thereby improving treatment efficacy and reducing adverse drug reactions. Continued advances in genomic technologies, artificial intelligence, and precision medicine, together with broader clinical implementation, are expected to make pharmacogenomics an integral component of personalized healthcare and improve patient outcomes worldwide.

References

1. Waldman, S. A., & Terzic, A. (2017). Clinical Pharmacology & Therapeutics: Past, Present, and Future. *Clinical pharmacology and therapeutics*, 101(3), 300–303. <https://doi.org/10.1002/cpt.592>
2. Kalow, W. (2005). Pharmacogenomics. In: Innocenti, F. (eds) *Pharmacogenomics. Methods in Molecular Biology™*, vol 311. Humana Press. <https://doi.org/10.1385/1-59259-957-5:003>
3. Qahwaji, R., Ashankyty, I., Sannan, N. S., Hazzazi, M. S., Basabrain, A. A., & Mobashir, M. (2024). Pharmacogenomics: A Genetic Approach to Drug Development and Therapy. *Pharmaceuticals (Basel, Switzerland)*, 17(7), 940. <https://doi.org/10.3390/ph17070940>
4. Zhou, Y., & Lauschke, V. M. (2022). The genetic landscape of major drug metabolizing cytochrome P450 genes—an updated analysis of population-scale sequencing data. *The pharmacogenomics journal*, 22(5-6), 284–293. <https://doi.org/10.1038/s41397-022-00288-2>
5. Weinshilboum, R. M., & Wang, L. (2017). Pharmacogenomics: Precision Medicine and Drug Response. *Mayo Clinic proceedings*, 92(11), 1711–1722. <https://doi.org/10.1016/j.mayocp.2017.09.001>
6. Sadee, W., Wang, D., Hartmann, K., & Toland, A. E. (2023). Pharmacogenomics: Driving Personalized Medicine. *Pharmacological reviews*, 75(4), 789–814. <https://doi.org/10.1124/pharmrev.122.000810>
7. Ali, U., & Mahato, R. K. (2026). Integration of pharmacogenomics into precision medicine: transforming healthcare for the next generation. *Annals of medicine and surgery (2012)*, 88(2), 2192–2193. <https://doi.org/10.1097/MS9.0000000000004691>
8. Lauschke, V. M., & Ingelman-Sundberg, M. (2026). The evolving landscape of pharmacogenomics: Current achievements and future directions. *Pharmacological reviews*, 78(2), 100122. <https://doi.org/10.1016/j.pharmr.2026.100122>